

Multiple Choice (10 marks)

1. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan
2. Can we build a secure PRP from a secure PRG?
 - (a) No, it cannot be done
 - (b) Yes, just plug the GGM PRF into the Luby-Rackoff theorem
 - (c) It depends on the underlying PRG
 - (d) Option text
3. Which of the following are most vulnerable to injection attacks?
 - (a) Session IDs
 - (b) Registry keys
 - (c) Network communications
 - (d) SQL queries based on user input
4. Why apps developed in languages like C, C++ is prone to Buffer-overflow?
 - (a) No string boundary checks in predefined functions
 - (b) No storage check in the external memory
 - (c) No processing power check
 - (d) No database check
5. Which of the following is not a block cipher operating mode?
 - (a) ECB
 - (b) CFB
 - (c) CBF
 - (d) CBC

True / False (10 marks)

Answer True or False

6. True or False: A buffer is a sequential segment of memory allocated for storing data, such as character strings or arrays of integers.
7. True or False: Blackbox fuzzers can effectively cover every line of code in a program without access to its internal structure.
8. True or False: Buffer-overflow vulnerabilities may persist in applications if boundary checks are not properly implemented.
9. True or False: Encapsulating Security Payload (ESP) is a component of the Secure IP Protocol used to provide confidentiality and integrity in network communications.
10. True or False: Nmap determines port states as open, filtered, and unfiltered during its scanning process.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the classic security properties in cybersecurity and critically analyze why availability is often considered distinct from these fundamental principles.
12. Discuss the role of authorization in cybersecurity, focusing on how it restricts user access to operations and data, and analyze its importance in protecting sensitive information.

End of Paper